

B.A. / B.Sc. (SIXTH SEMESTER) EXAMINATION, 2025

MATHEMATICS

MATH 312— Vector Calculus

TIME ALLOWED : THREE HOURS

Maximum Marks— 60

*Attempt five questions in all, selecting one question from each Unit.
All questions carry equal marks.*

Unit-I

Q1 (a) Define the following with suitable example: [4]

- (i) Vector and scalar.
- (ii) Unit vector
- (iii) Collinear vector
- (iv) Equal vector

(b) Write geometric interpretation of [4]

$$|u_1 + u_2|^2 + |u_1 - u_2|^2 = 2[|u_1|^2 + |u_2|^2].$$

(c) Find the unit vector in the direction of **sum** and **difference** of the vectors $u = 2i + j - 5k$ and $v = 2i + j + 3k$ [4]

Q2 (a) Write examples of two different vectors u and v such that [4]

- (i) u and v have the same magnitude.
- (ii) u and v have the same direction.

(b) If $u = 5i - j - 3k$ and $v = i + 3j - 5k$, then show that the vectors $u + v$ and $u - v$ are perpendicular. [4]

(c) Find the area of the parallelogram whose adjacent sides are given by the vectors $u = 5i + 2j + 2k$ and $v = 2i - 5j + 7k$. [4]

Unit-II

Turn over

- Q3** (a) If r is the differentiable vector-valued function of ' t ' of constant length, then prove that $r \cdot \frac{dr}{dt} = 0$ [4]
- (b) Define the vector product of four vectors. If u_1, u_2, u_3 and u_4 be any four vectors, then prove that [4]
- $$(u_1 \times u_2) \times (u_3 \times u_4) = [u_1 \ u_2 \ u_3]u_4 - [u_1 \ u_2 \ u_4]u_3.$$
- (c) If $u(t) = i \sin^2 t - 2j \cos 3t + k t^3$, then find $\lim_{t \rightarrow \frac{\pi}{4}} u(t)$. [4]
- Q4** (a) Define scalar product of four vectors. If $u_1 = i + 3j - 2k$, $u_2 = -2i + 5k$, $u_3 = 7i - 5j + 29k$, and $u_4 = i - 7j - 4k$ be four vectors, then find the $(u_1 \times u_2) \cdot (u_3 \times u_4)$. [4]
- (b) Given $F(t) = (4t^2 - 3t)i + 8t^3j - 9tk$, then evaluate $\int_{t=3}^{t=9} F(t)dt$. [4]
- (c) A particle moves along the curve $x = 2t^2, y = t^2 - 4t, z = 2t - 5$, where t is the time. Find the components of its velocity and acceleration at time $t = 1$ in the direction of $i - 3j + 2k$. [4]

Unit-III

- Q5** (a) A fluid motion is given by $V = (y+z)\hat{i} + (z+x)\hat{j} + (x+y)\hat{k}$. [4]
- (i) Is the motion irrotational? If so, find the velocity potential.
- (ii) Is the motion possible for incompressible fluid?
- (b) Prove that a scalar valued function is constant if and only if $\text{grad } f = 0$. [4]
- (c) Prove that $\text{div} \left(\frac{\vec{r}}{r^3} \right) = 0$, where $\vec{r} = xi + yj + zk$ and $r = \sqrt{x^2 + y^2 + z^2}$. [4]
- Q6** (a) If A and B are irrotational, then prove that $A \times B$ is solenoidal. [4]
- (b) Find the first order derivatives of $w = x^2y + \log xy$ using chain rule, where $x = 2t^2 + t; y = \frac{1}{t^2}$. [4]

- (c) If $u = x + y + z$, $v = x^2 + y^2 + z^2$, $w = yz + zx + xy$, then prove that $(\text{grad } u) \cdot \{(\text{grad } v) \times (\text{grad } w)\} = 0$. [4]

Unit-IV

- Q7 (a) Find the directional derivative of $f(x, y, z) = x^2 - 2y^2 + 4z^2$ at the point $(1, 1, -1)$ in the direction of the vector $2i + j - k$. Also find the direction of the maximum directional at $(1, 1, -1)$ and its maximum value. [4]
- (b) Find the flux across the surface of the parabolic cylinder $y^2 = 8x$ in the first octant bounded by the planes $y = 2$ and $z = 4$, when the velocity vector $V = 3y i - zj + x^2 k$. [4]
- (c) Find the equations of the tangent plane and normal line to the surface $xyz = 4$ at the point $(1, 2, 2)$. [4]
- Q8 (a) Find the surface area of the plane $x + 2y + 2z = 12$ cut off by $x = 0$, $y = 0$, and $x^2 - y^2 = 16$. [4]
- (b) Compute the line integral $\int_C (y^2 dx - x^2 dy)$ about the triangle whose vertices are $(1, 0)$, $(0, 1)$ and $(0, 1)$. [4]
- (c) Evaluate $\oint_S F \cdot \vec{N} ds$ where $F = 3x^2 y i - 2y^2 j + 4xz^2 k$ and S is the closed surface of the region in the first octant bounded by the cylinder $y^2 + z^2 = 9$ and the planes $x = 0$, $x = 2$, $y = 0$, $z = 0$. [4]

Unit-V

- Q9 (a) State and prove Stoke's theorem. [6]
- (b) Verify Green's theorem for $\oint_C [(xy + x^2)dx + (x^2 + y^2)dy]$, where C is the curve bounded by the lines $y = \pm 1$ and $x = \pm 1$. [6]
- Q10 (a) State and prove Gauss theorem. [6]
- (b) Verify Stoke's theorem for $F = (x^2 + y^2) i - 2xy j$ taken around the rectangle bounded by the lines $x = \pm a$, $y = 0$, $y = b$. [6]

B.A. / B.Sc. (SIXTH SEMESTER) EXAMINATION, 2024

MATHEMATICS

MATH 312 — Vector Calculus

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UNIT I

- 1.(a) If $u = (3, -2)$ and $v = (-2, 5)$, then find component form and length of the following vectors:
- $2u - 3v$.
 - $\frac{3}{5}u + \frac{4}{5}v$.
- (b) A force of 6 newtons is applied in the direction of the vector $v = 2i + 2j - k$. Express the force F as a product of its magnitude and direction.
- (c) Define the followings with suitable example:
- Vector and scalar
 - Cross product of vectors, when angle is not given.
- 2.(a) If u, v , and w are any vectors and r, s are scalars, then prove that
- $v \times u = -(u \times v)$
 - $u \times (v \times w) = (u \cdot w)v - (u \cdot v)w$
- (b) Find a unit vector perpendicular to the plane of $P(1, -1, 0)$, $Q(2, 1, -1)$, and $R(-1, 1, 2)$.
- (c) If either vector $a = 0$ or $b = 0$, then $a \cdot b = 0$. But the converse need not be true. Justify your answer with an example.

UNIT II

- 3.(a) Given $F(t) = (5t^2 - 3t)i + 6t^3j - 7tk$, then evaluate $\int_{t=2}^{t=4} F(t)dt$.
- (b) Let $a = 5i + 6j + 7k$; $b = 2i + 3j + 4k$; $c = i + 2j + 3k$ and $d = 3i + j + 5k$. Then find $(a \times b) \cdot (c \times d)$.
- (c) Show that $f(t) = \sin t i + \cos t j + \sqrt{3} k$ has constant length and is orthogonal to its derivative.
- 4.(a) Define the following:
- vector product of four vectors.
 - scalar product of three vectors.
- (b) Let $a = i - j + k$; $b = 2i + j + k$; $c = i + j - 2k$. Then by using the scalar triple product of vectors, verify that $[a b c] = [b c a] = -[a c b]$.
- (c) If $f(t) = \cos t i + \sin t j + tk$, then find $\lim_{t \rightarrow \frac{\pi}{4}} f(t)$.

PTC

UNIT III

- 5.(a) If $u = f(y - z, z - x, x - y)$, then prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.
- (b) Show that each of the following vectors are solenoidal
- $(x + 3y)i + (y - 3z)j + (x - 2z)k$,
 - $3y^4z^2i + 4x^3z^2j + 3x^2y^2k$.
- 6.(a) Show that $\nabla^2 f(r) = f''(r) + \frac{2}{r} f'(r)$, where $r = xi + yj + zk$ and ∇ is del operator.
- (b) Write the short note on the followings:
- Physical interpretation of divergence
 - Physical interpretation of curl.

UNIT IV

- 7.(a) What is the directional derivative of $u = xy^2 + yz^3$ at the point $(2, -1, 1)$ in the direction of normal to the surface $x \log z - y^2 = -4$ at the point $(-1, 2, 1)$?
- (b) Find the equation for the tangent plane and normal line of the level surface $3x^2 + 5y^2 + z^2 = 9$ at the point $(2, 3, 6)$.
- 8.(a) If velocity vector is $F = yi + 2j + xz k$ m/sec., then show that the flux of water through the parabolic cylinder $y = x^2$, $0 \leq x \leq 3$, $0 \leq z \leq 2$ is 69 m³/sec.
- (b) A vector field is given by $F = (\sin y)i + x(1 + \cos y)j$. Evaluate the line integral over a circular path given by $x^2 + y^2 = a^2$, $z = 0$.

UNIT V

- 9.(a) State and prove Gauss divergence theorem.
- (b) If ϕ is a solution of the Laplace's equation, then prove that $\nabla\phi$ is both solenoidal and irrotational.
- 10.(a) Verify Gauss divergence theorem for $F = 4xz i - y^2 j + yz k$ taken over the cube bounded by $x = 0, x = 1; y = 0, y = 1; z = 0, z = 1$.
- (b) State and prove Green's theorem in a plane.

**B.A. / B.Sc. (SIXTH SEMESTER)
EXAMINATION, 2023**

MATHEMATICS

MATH 312— Vector Calculus

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UNIT I

1. (a) What is vector? Explain with suitable example. If $v = 2i - 3j + 4k$ is the vector with initial point $(6, 2, 1)$, then find the terminal point. 4
- (b) Let $u = 0i - 0j + 0k$ and $v = i + 0j + 0k$ be two vectors in $\mathbb{R}^3$. Then what can you say about magnitude and direction of the vectors u and v ? Also discuss the magnitude and direction of the resultant. 4
- (c) Let $u = (u_1, u_2, u_3)$ and $v = (v_1, v_2, v_3)$ be two vectors in $\mathbb{R}^3$. Then find its dot product and cross product. Let

Turn over

$$\int_{t=2}^{t=4} F(t) dt. \quad 4$$

- (c) Define the vector product of four vectors. If u_1, u_2, u_3 and u_4 are any four vectors, then prove that :

$$(u_1 \times u_2) \times (u_3 \times u_4) = [u_1 u_2 u_4] u_3 - [u_1 u_2 u_3] u_4. \quad 4$$

UNIT III

5. (a) A fluid motion is given by :

$$V = (y + z)\hat{i} + (z + x)\hat{j} + (x + y)\hat{k}.$$

- (i) Is the motion irrotational? If so, find the velocity potential.

- (ii) Is the motion possible for incompressible fluid? 4

- (b) Write short note on physical interpretation of curl. Also prove that if u and v are irrotational, then $u \times v$ is solenoidal. 4

- (c) Write a formula to determine the area of the parallelogram. Also find the area of a parallelogram whose adjacent sides are represented by the vectors $u = 2i - 2j + 5k$ and $v = i + 2j + 4k$. 4

6. (a) Define the divergence of vector point function. If $R = xi + yj + zk$, then show that :

(i) $\nabla \cdot R = 3$

(ii) $\nabla \times R = 0$. 4

- (b) Find the first order derivatives of

$$w = x^2 + y^2 + xy + \log xy$$

using chain rule, where $x = t^2 + t$; $y = \frac{1}{t^2}$. 4

UNIT V

9. (a) State and prove Stoke's theorem. 6
- (b) Verify Green's theorem for $\oint_C [(xy + y^2) dx + x^2 dy]$, where C is bounded by $y = x$ and $y = x^2$. 6
10. (a) State and prove Green's theorem. 6
- (b) Verify Stoke's theorem for $F = (x^2 + y^2)i - 2xyj$ taken around the rectangle bounded by the lines $x = \pm a$, $y = 0$, $y = b$. 6